

Unreal Engine 5.6 Simulator Doc

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Startup:

Creating New Unreal Engine Project

Game → *Blank* → Set as C++ & Scalable

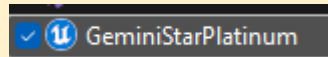
Installing Visual Studio

For installation workloads, choose *.NET desktop development*, *Desktop development with C++*, and *Game development with C++*.

Getting Visual Studio Integration Tool Plugin

You may be required by the project to install the *Visual Studio Integration Tool*, but you can't

install it with *Unreal Engine 5.6*.



Go to the project file and set the *Visual Studio Integration Tool* plugin to *false*. After opening the project, you can create a *Visual Studio* project through the *Tools* tab. From within *Visual Studio*, you can just install the *Visual Studio Integration Tool Plugin* for the engine. You could probably find a way without the plugin, but it doesn't hurt. You will need to put the plugin into a folder named *Marketplace*, which you may need to create yourself → *C:\Program Files\Epic Games\UE_5.6\Engine\Plugins\Marketplace*.

In Editor:

Fast FPS Tip



Set to *Unlit* for instant FPS boost.



Otherwise, for decent visuals, go to *Viewport Scalability*, and set *Global Illumination* to *Medium*.



Hide or delete *Volumetric Clouds*. They take up so much rendering.

Project Settings

Substepping: Enabled

Max Substep Delta Time: 0.0013

Max Substeps: 16

Position Iterations: 50 // Doesn't need to be nearly this high, but CPU isn't a big issue

Velocity Iterations: 30

Projection Iterations: 30

Enable Enhanced Determinism: Enabled

Enable Stabilization: Enabled

Anti-Aliasing Method: Temporal Super-Resolution (TSR)

Creating New C++ Class

Tools → New C++ Class...

Importing glTF Models

Content Drawer → Content folder → Click *Import*

→ Select the glTF file in the folder with the bin file

You can do default import options. If you want to disable collisions, which is what I did, turn *Import Collisions* off. If you want to combine all parts into a single static mesh, select *Combine Static Meshes*. Consider using the fields in *Common* for folder management and offsetting the origin point if desired.

Setting Collisions Manually

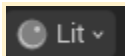
Content Drawer → Double click or *Edit* static mesh → Click the *Collision* dropdown menu → *Add Box Simplified Collision* → In the *Details* window, *Primitives* → *Boxes* → *Index [0]* → Adjust parameters as desired

Debugging Telescope Model

If the telescope is not adhering to physics or oddly unbalanced, you will need to manually set the reference frames. This is essential to adjust for the appropriate physics and not follow what is auto-generated from Unreal Engine.

Nanite

This project uses *Nanite*, which is similar to hierarchical levels of detail (HLOD). It is super powerful and customizable, so play around with it. On a normal computer, like my laptop, the engine should run at 120+ fps, but the office computers are quite slow, so it can save you a lot of frames.



You can also select *Nanite Visualization* for a good understanding of how it works.

Importing LIDAR From OpenTopography

▶ High Resolution Heightmap Data - Building Worlds In Unreal - Episode 16

Make sure you import to *TerreSculptor* as *GeoTIFF raster imagery*, not as *TIF image*. There are hundreds of different ways to do this, but I followed this tutorial generally.

Model Information:

Misaligned Models?

The models should point toward the X direction. The telescope and dome point toward X but have a different pitch and yaw due to modeling difficulties. In fact, if you try to change the origin in *Unreal Engine*, you will likely freeze your engine for a few hours. **Be careful and save beforehand.**

Direction Cheat Sheet

Yaw = Around Z = XY plane = Swing1

Pitch = Around Y = XZ plane = Swing2

Roll = Around X = YZ plane = Twist

Unreal Engine loves to mix everything and the models might not align the same direction!

Blueprint Widget

📺 Unreal-5: UI-Menu Display & Mouse Visibility!